

COMPUTER SCIENCE BSC CURRICULUM

Program: (English for Foreign Students)

PREPARATORY SEMESTER CURRICULUM

ASSESSMENT TYPES:

P: Practice Grade | XP: Mixed with Practice

PREPARATORY SEMESTER (25 Credits)

IP-18fPRECPEG: Computers in practice

0 Lecture, 0 Practice, 2 Laboratory, 0 Consultation | XP | 2 Credits

IP-18fPRESPROGEG: Scratch Programming

0 Lecture, 0 Practice, 4 Laboratory, 0 Consultation | XP | 4 Credits

IP-18fPRETMKG: Learning Methodology

0 Lecture, 2 Practice, 0 Laboratory, 0 Consultation | P | 1 Credit

IP-18fPRETMG: Topics in Mathematics

0 Lecture, 8 Practice, 0 Laboratory, 0 Consultation | P | 8 Credits

IP-18fPREFICTG: Introduction to computational thinking

0 Lecture, 0 Practice, 4 Laboratory, 0 Consultation | P | 4 Credits

IP-18fPREIMEG: Intermediate English

0 Lecture, 6 Practice, 0 Laboratory, 0 Consultation | P | 6 Credits

CREDIT SUMMARY:

Total Preparatory Semester Credits: 25

ASSESSMENT TYPES:

XCP: Mixed with Continuous Assessment | XE: Mixed with Exam | XP: Mixed with Practice

CP: Continuous Assessment | P: Practice Grade | E: Exam

SEMESTER 1 (29 Credits)

IP-18fSZGREG: Computer systems

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XCP | 5 Credits

IP-18fPROGEG: Programming

2 Lecture, 0 Practice, 3 Laboratory, 1 Consultation | XCP | 6 Credits

IP-18fIMPROGEG: Imperative programming

2 Lecture, 0 Practice, 3 Laboratory, 0 Consultation | XCP | 5 Credits

IP-18fFUNPEG: Functional programming

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XE | 5 Credits

IP-18fMATAG: Basic Mathematics

0 Lecture, 4 Practice, 0 Laboratory, 0 Consultation | CP | 4 Credits

IP-18fTMKG: Learning Methodology

0 Lecture, 1 Practice, 0 Laboratory, 0 Consultation | P | 1 Credit

IP-18fIVMEG: Business fundamentals

1 Lecture, 2 Practice, 0 Laboratory, 0 Consultation | XP | 3 Credits

SEMESTER 2 (30 Credits)

IP-18fPNYEG: Programming languages

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XCP | 6 Credits | Prerequisite: IP-18fIMPROGEG

IP-18fOEPROGEG: Object-oriented programming

2 Lecture, 0 Practice, 3 Laboratory, 1 Consultation | XCP | 6 Credits | Prerequisite: IP-18fPROGEG

IP-18fWF1EG: Web development

1 Lecture, 0 Practice, 2 Laboratory, 0 Consultation | XCP | 3 Credits | Prerequisite: IP-18fSZGREG (weak)

IP-18fAA1E: Algorithms and data structures I (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fAA1G (weak)

IP-18fAA1G: Algorithms and data structures I (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fMATAG, IP-18fIMPROGEG

IP-18fDM1E: Discrete mathematics I (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fDM1G (weak)

IP-18fDM1G: Discrete mathematics I (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | CP | 3 Credits | Prerequisite: IP-18fMATAG

IP-18fAN1E: Analysis I (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fAN1G (weak)

IP-18fAN1G: Analysis I (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fMATAG

SEMESTER 3 (22 Credits)**IP-18fAA2E: Algorithms and data structures II (Lecture)**

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fAA2G (weak)

IP-18fAA2G: Algorithms and data structures II (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fAA1E

IP-18fWPEG: Web programming

1 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XCP | 4 Credits | Prerequisite: IP-18fWF1EG

IP-18fPROGTEG: Programming technology

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XCP | 5 Credits | Prerequisite: IP-18fOEPROGEG

IP-18fAN2E: Analysis II (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fAN2G (weak)

IP-18fAN2G: Analysis II (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fAN1E

IP-18fDMAG: Application of discrete models

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fDM1E

SEMESTER 4 (22 Credits)**IP-18fOPREG: Operating systems**

1 Lecture, 0 Practice, 1 Laboratory, 1 Consultation | XP | 3 Credits | Prerequisite: IP-18fSZGREG

IP-18fAB1E: Databases I (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fAB1G (weak)

IP-18fAB1G: Databases I (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 0 Consultation | P | 2 Credits | Prerequisite: IP-18fAA1E

IP-18fSZTEG: Software technology

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XCP | 5 Credits | Prerequisite: IP-18fPROGTEG

IP-18fSZA1E: Fundamentals of theory of computation I (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fSZA1G (weak)

IP-18fSZA1G: Fundamentals of theory of computation I (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fDM1E

IP-18fNM1E: Numerical methods (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fNM1G (weak)

IP-18fNM1G: Numerical methods (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fAN2E

SEMESTER 5 (24 Credits)**IP-18fKPROGEG: Concurrent programming**

1 Lecture, 0 Practice, 1 Laboratory, 1 Consultation | XCP | 3 Credits | Prerequisite: IP-18fPNYEG

IP-18fTKHE: Telecommunication networks (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fTKHG (weak)

IP-18fTKHG: Telecommunication networks (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fOEPROGEG

IP-18fMIAE: Artificial intelligence

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits

IP-18fAB2E: Databases II (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fAB2G (weak)

IP-18fAB2G: Databases II (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fAB1E

IP-18fSZA2E: Fundamentals of theory of computation II (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fSZA2G (weak)

IP-18fSZA2G: Fundamentals of theory of computation II (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fSZA1E

IP-18fVSZG: Probability and statistics

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fAN2E

SEMESTER 6 (20 Credits)**IP-24fSZD: Thesis consultation**

Prerequisite: Handling in the Thesis Topic Declaration

+You need to complete 23 credits from the following Compulsory Elective courses:

Compulsory Elective Courses

--- Recommended SEMESTER 2 ELECTIVES ---

IP-24fKVSZKBIZTE: Cybersecurity Basic

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits | Prerequisite: IP-18fWF1EG (weak)

--- Recommended SEMESTER 3 ELECTIVES ---

IP-18fKVBGTE: Introduction to machine learning

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IP-18fMATAG

IP-18fKVPREE: Programming theory (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fKVFP (weak)

IP-18fKVPREG: Programming theory (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fMATAG

IP-18fKVPYEG: Python

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XE | 5 Credits

IP-18fKVIADSE: Applied Data Science

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits

IP-24fKVSZGE: Computer graphics (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IP-24fKVSZGG (weak)

IP-24fKVSZGG: Computer graphics (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fMATAG, IP-18fIMPROGEG

--- Recommended SEMESTER 4 ELECTIVES ---

IP-18fKVKBRE: Cryptography and security (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fKVKBRE (weak)

IP-18fKVKBRE: Cryptography and security (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fDMAG

IP-18fKVPYEG: Python

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XE | 5 Credits

IP-24fKVKSZGG: Intermediate Computer Graphics

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-24fKVSZGG

IP-24fKVNFAEG: Native cloud computing applications

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XE | 5 Credits | Prerequisite: IP-18fKVPYEG

IP-24fKVHWPEG: Advanced web programming

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XPG | 5 Credits | Prerequisite: IP-18fWPEG

--- Recommended SEMESTER 5 ELECTIVES ---

IP-18fKVPREE: Programming theory (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fKVFP (weak)

IP-18fKVPREG: Programming theory (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fMATAG

IP-18fKVPJG: Tools of software projects

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | CP | 3 Credits | Prerequisite: IP-18fPNYEG

IP-18fKVFP: Compilers (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IP-18fKVFP (weak)

IP-18fKVFP: Compilers (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 0 Consultation | P | 2 Credits | Prerequisite: IP-18fOEPGEG

IP-18fKVADA: ADA

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XP | 5 Credits | Prerequisite: IP-18fIMPROGEG

IP-18fKVPYEG: Python

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XE | 5 Credits

IP-18fKVIADSE: Applied Data Science

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits

IP-24fKVSZGE: Computer graphics (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IP-24fKVSZGG (weak)

IP-24fKVSZGG: Computer graphics (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fMATAG, IP-18fIMPROGEG

IP-24fKVNFAEG: Native cloud computing applications

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XE | 5 Credits | Prerequisite: IP-18fKVPYEG

--- Recommended SEMESTER 6 ELECTIVES ---

IP-18fKVKRBE: Cryptography and security (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IP-18fKVKRBG (weak)

IP-18fKVKRBG: Cryptography and security (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-18fDMAG

IP-18fKVADA: ADA

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XP | 5 Credits | Prerequisite: IP-18fIMPROGEG

IP-18fKVPYEG: Python

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XE | 5 Credits

IP-24fKVKHSZGG: Intermediate Computer Graphics

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | P | 3 Credits | Prerequisite: IP-24fKVSZGG

IP-24fKVNFAEG: Native cloud computing applications

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XE | 5 Credits | Prerequisite: IP-18fKVPYEG

IP-24fKVHWPEG: Advanced web programming

2 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XPG | 5 Credits | Prerequisite: IP-18fWPEG

CREDIT SUMMARY

Compulsory Courses: 127 Credits

Compulsory Electives: 23 Credits

Electives: 10 Credits

Thesis: 20 Credits

TOTAL DEGREE CREDITS: 180 Credits

You need to complete at least 10 credits from other elective courses, please read the „Elective Courses” section of the website for more information about this. (You can also choose to include some of the above „compulsory elective” courses in your elective credits.) If you add up all the credit numbers, you will see that you have to complete 180 credits altogether during your studies.